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	The intricate process of embryonic development initiates with a single totipotent stem cell that by undergoing multiple cycles of division/differentiation gives rise to an entire multicellular organism. Until recently it was thought that with each subsequent division, the progeny of these original stem cells shed some of their totipotency and began committing to a certain tissue and/or cellular lineage. Recent studies have shown that multipotent stem cells persist throughout life and participate in normal replacement and repair within their respective tissues. Furthermore, results from both in vitro and in vivo studies have now suggested that stromal/mesenchymal stem cells from bone marrow (BM-MSC) may harbor the ability to de-differentiate and reprogram themselves to give rise to cells of unrelated lineages. We have recently begun to use our laboratory's unique human-sheep xenograft model system of in utero human stem cell transplantation to explore the plasticity that appears to exist within the primitive stem cell population present within several organs. We have now shown that human neural stem cells are able to produce both differentiated blood cells and hematopoietic stem/progenitor cells with the ability to engraft the hematopoietic systems of secondary transplant		

Task Description:	recipients. We have also demonstrated the formation of human hepatocytes following hematopoietic stem cell, mesenchymal stem cell, or neural stem cell transplantation in utero. Since blood, brain, and liver have not yet been shown to be capable of interconverting in vitro, we concluded that the influence of the fetal microenvironment was responsible for unlocking the hidden potential of these organ-specific stem cells, enabling them to trans-differentiate and give rise to cells of other tissues. Indeed, the fetus represents a unique system in which to assay stem cell differentiation potential and plasticity because of both the wide array of inductive environmental signals and the need for multiple cell types created by the exponential growth rate of a developing fetus. However, based on the rapidly accumulating data that conditions of microgravity alter the gene expression of cultured cells, we wish in the present proposal to explore an alternate explanation for the plasticity of the stem cell(s) within our model system. In addition to the afore-mentioned characteristics, the fetus also represents a unique biological system because it is suspended in a fluid-filled sac and thus approaches weightlessness, at least at the early stages of development. Thus, we propose that the weightless state of the early gestational fetus is also a contributing factor to the observed plasticity within the stem cell compartment of various organ systems. While a fetal transplantation model system is ideal for evaluating many aspects of stem cell plasticity, the ability to address these issues in vitro would allow dissection and manipulation of the processes controlling the interconversion of cells from seemingly disparate embryologic origins. Unfortunately, to date, it has not been possible to perform studies such as these in vitro due to the difficulties with accurately recreating each organ's microenvironment in traditional monolayer culture systems. The development of NASA's rotating wall vessel (RWV) has now allowed the successful long-term culture of both mixed cell populations that have been largely refractory to standard culture approaches and relatively large tissue fragments from a variety of organs, maintaining normal cytoarchitecture. Thus, the overall goal of these studies is to use the RWV to begin addressing the complex question of mesenchymal stem cell plasticity in vitro while simultaneously evaluating the impact that gravity (or its relative absence) has on the inherent plasticity of stem cells and their expression of the gene pathways responsible for this malleability. The use of the RWV will allow the performance of carefully designed in vitro studies employing clonally derived retroviral marked populations of BM-MSC that will enable us to answer the difficult question of whether these stem cells constitute a mixed population, each member of which is committed to its own embryonic fate, or whether these stem cells are truly pluripotent. We will perform studies in which 1) BM-MSC are cultured alone under conditions designed to induce differentiation into cells of liver, muscle, bone, cartilage, endothelial, epithelial, and neural derivation, 2) BM-MSC are cultured in each of these conditions in the presence of cellular suspensions prepared from each of the afore-mentioned tissues, and 3) BM-MSC are cultured in each of these conditions in the presence of tissue fragments from each of the afore-mentioned organs. By including studies with BM-MSC alone, with mixed cell types, and with tissue fragments, we will be able to examine BM-MSC plasticity in vitro at the single cell, multi-cell (spheroid), and tissue (organoid) levels. The studies at the single cell level will also address the important issue of the impact of microgravity on cellular proliferation and telomere/telomerase homeostasis. Once these in vitro studies have been completed, we will then use our unique human-to-sheep model of in utero transplantation to evaluate whether culture under conditions of microgravity impacts on the in vivo engraftment ability and plasticity/differentiative potential of BM-MSC. MSC may play a role in normal cellular regeneration and, more importantly, be a quiescent population of cells that can function as a rapid source of cells to provide specific organ cells in case of injury. If microgravity affects the ability of MSC to differentiate into other cell types, thus preventing normal regeneration/repair within the organ, this may affect the health of humans subjected to conditions of imponderability. If we are able to detect which are the specific differentiation pathways that are altered under microgravity conditions, it may be possible to develop strategies to avoid this problem. It is known that microgravity affects several organs and systems in a negative way, for instance, bone loss, immunosuppression, re-activation of endogenous viral infections, etc. Most of the studies performed so far have studied the effects of microgravity on mature cells, for instance, alterations in monocytic or lymphocytic physiology. In a similar way, osteoporosis has been studied at the level of bone cells such as osteoblasts. Since it is well known that tissue-specific stem cells persist throughout the adult life and have a critical role in tissue regeneration and repair, it is obvious that the study of the effects of microgravity on tissue-specific stem cells can open a new understanding of the effects of weightlessness on human life.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies have shown that the study of stem cells subjected to microgravity conditions in culture has led to pertinent answers to physiologic alterations suffered by astronauts subjected to space-flight conditions. Thus, if the mechanisms leading to these alterations are known, it will be possible to find countermeasures to re-establish normal homeostasis. Furthermore, lessons learned from the alterations induced by microgravity in stem cells' differentiative pathways can contribute to the understanding of stem cell biology and lead to the development of new stem cell-based therapies on earth.
	We have previously reported that Stro-1 is able to identify populations of human Mesenchymal Stem Cells (MSCs) in different organs. Here, we investigated whether microgravity (microG) can alter the "stemness" of MSCs in such a way as to hinder the ability of these cell populations to differentiate into various cells/lineages. As the effects of microG on bone loss are well known, we investigated whether alterations in MSC's differentiative pathways could lead to osteoporosis. Thus, we compared the ability of MSCs to differentiate into adipocytes and bone under regular or microG culture conditions. We used Ca2+ content, Von Kossa staining and tetracycline incorporation to determine osteocytic and Oil-Red-O staining for adipocytic differentiation, respectively. Adipocytes and osteocytes were obtained within 9 and 12 days, in regular cultures. Under microG conditions, adipocytic differentiation occurred in 12 days, similar to that of regular cultures. However, osteocytic differentiation under ?G conditions took 4 times longer than cultures grown under normal G. Since MSCs give rise to both osteoblasts and adipocytes, it is possible that an imbalance in these differentiative pathways may lead to impaired osteogenesis, contributing to osteoporosis. Furthermore we have shown that Mesenchymal stem cells (MSC) are able upon transplantation to differentiate into specialized cells of multiple organs. Based on these findings, we hypothesized that MSC may be responsible, in part, for maintaining the normal homeostasis/replacement of cells within these different organs. We have previously shown that microgravity contributes to an imbalance of MSC's differentiative pathways towards adipogenesis over osteogenesis in vitro, possibly leading to osteoporosis. In the present studies, we investigated whether microgravity may also alter the differentiative potential of MSCs in such a way as to hinder the ability of MSC to differentiate into other different cells/lineages. To accomplish this aim, adult bone marrow-derived MSC (BM MSC), fetal liver MSC (FL MSC), or brain MSC (BR MSC) were seeded onto cytodex 3 beads and grown in a rotary cell culture system for 5 days, and subsequently transplanted at a concentration of 1 million cells/fetus into 12 fetal sheep recipients. Flow cytometric evaluation of the recipients at 60 days post-transplant for the presence of donor (human)-derived blood and BM revealed that all MSC populations

Task Progress:	generated multilineage hematopoietic cells, including CD34+ cells, at differing levels. However, when the levels of human cell engraftment of these animals were compared with control sheep transplanted with the same MSCs grown in regular cultures, it became apparent that MSCs grown under microgravity conditions generated multilineage blood cells at a lower level than their regular culture counterpart, suggesting that microgravity impairs either MSCs engraftment and/or in vivo differentiation. Other organs such as liver and pancreas were also evaluated for the presence of human organ-specific cells to investigate if the differentiative potential of MSC cells grown in microgravity is equivalent to that of MSCs cultured under conditions of normal gravity. We found that differentiation towards a hepatocyte-like phenotype was impaired in MSCs grown under microgravity conditions. MSCs grown in normal gravity conditions generate, upon transplantation, 5-10% of human hepatocytes, while the animals transplanted with MSCs grown in microgravity very rarely contained human hepatocytes in the livers. In contrast, most of the sheep transplanted with MSC grown under microgravity conditions contained human beta cells within the Islets of Langerhans, and these cells produced human insulin, while animals transplanted with MSCs in grown in regular cultures did not. These studies may facilitate the understanding of the significant changes of plasma insulin and glucose levels observed in astronauts during space flights. In order to start dissecting mechanisms by which these cells have been reprogrammed to chose one differentiative pathway versus another, proteomic analysis of MSC cells grown under microgravity and normal conditions were performed and Genomic analysis is currently underway. Figure 10 shows a 2 -D Polyacrylamide gel electrophoresis of liver MSC grown either at 1G or microgravity conditions. Of interest is that serum albumin-precursor and serum albumin characteristically produced by hepatocytes were down-regulated in MSC upon exposure to microgravity supporting the results obtained in vivo which showed a lack of differentiation of MSC towards a hepatocytic phenotype. Also the fact that these MSC upon exposure to microgravity were producing Fetuin in excess may contribute to the f rapid kidney stone formation in astronauts on space travels, since fetuin is a serum glycoprotein in the cystatin family, associated with the regulation of soft tissue calcification that inhibits unwanted mineralization Furthermore, the downregulation of osteocalcin in cells grown under microgravity conditions also supports the data described above, in which MSCs express a deficiency in mineralization upon culture in microgravity.
Bibliography Type:	Description: (Last Updated: 08/28/2021)
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